

Dissolved Oxygen Sensor User Manual

Xi'an Desun Uniwill Electronic Technology Co., Ltd.

Please Read This Manual Carefully Before Use.



Preface

Thank you very much for choosing our company!

Please read this manual carefully before using this product, and save it for reference. Please observe the operating procedures and precautions in this manual.

To ensure that the after-sale protection provided by this instrument is effective, please do not use and maintain the instrument in any way other than specified in this manual.

As the failure to observe the precautions specified in this operation manual, any failures and losses caused are not covered by the manufacturer's warranty, and the manufacturer does not assume any related responsibility. Please keep all documents in a safe place. If in doubt, please contact our after-sales service department.

When receiving the instrument, please carefully open the package and check whether the instrument and accessories are damaged due to shipping. If damage is found, please contact Desun Uniwill after-sales service department and keep the packaging for return.

When the instrument fails, please do not repair it by yourself, please contact our after-sales service department.

I. User Notice

- ✓ Please read this manual carefully before use and save it for reference.
- ✓ Please follow the operating procedures and precautions in this manual.
- ✓ When receiving the instrument, please carefully open the package and check whether the instrument and accessories are damaged due to shipping. If any damage is found, please inform the manufacturer and distributor immediately, and keep the packaging for return.
- ✓ When the instrument malfunctions, do not repair it yourself. Please contact the after-sales department of the manufacturer directly.

II. Overview

Features

- 0-20mg/L or 0-200% saturation
- 0~50 °C; IP68 protection level
- No film, no electrolyte, no polarization
- No need to consume oxygen, no flow rate
- Built-in temperature sensor, automatic temperature compensation
- Free from chemical interferences such as sulfides
- Small annual drift, fast response, and more accurate measurement
- Free of maintenance, longer application cycle and lower cost of use
- Fluor cap replacement is simple
- Support RS-485, MODBUS protocol



Measuring Principle:

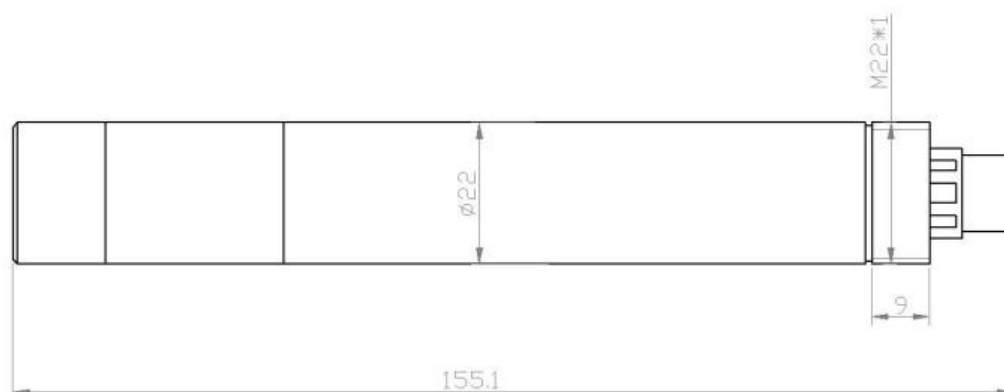
The fluorescent dissolved oxygen sensor is based on the quenching principle of the specific fluorescence of the specific substance in physics. Blue light from a light-emitting diode (LED) illuminates the fluorescent material on the inner surface of the fluorescent cap, and the fluorescent material on the inner surface is excited to emit red light by detecting the phase difference between red and blue light and the internal calibration value. The ratio is calculated to calculate the concentration of oxygen molecules, and the final value is automatically compensated by temperature and pressure.

Technical Parameters

Item	Parameters
Model	DS380
Principle	Fluorescence analysis method
Measuring range	0~20mg/L or 0-200% saturation
Response time	3 sec

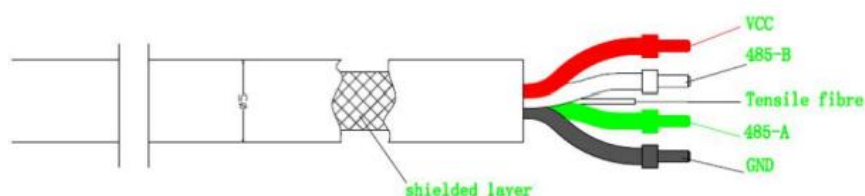
Accuracy	1%
Protection level	IP68
Installation	Submersible
Temperature sensor	NTC
Temperature range	0~50°C
Temperature accuracy	±0.2°C
Output	RS-485, MODBUS protocol
Power supply	DC 6~12V, current <50mA
Size	Φ26*155 mm
Probe cable length	Default 5 m
Fluorescent cap life	1 year
Maximum working pressure	3bar
product material	POM/SS316/Titanium
Calibration method	One or Two-point calibration
Power consumption	< 0.5 W

III. Dimensions:



IV. Cable Information

4 wire AWG-24 or AWG-26 shielding wire. OD=5mm



- Red line----power supply (VCC)
- White line---- 485 data -B (485-B)
- Yellow line---- 485 data -A (485-A)
- Black line----ground (GND)
- Bare line---- shielding layer

V. Installation Tips

- 1) Installation should try to avoid bubbles, sludge, etc. on the detection parts, as far as possible from the oxygenator, oxygenation pump, aeration tank installation.
- 2) When installing the optical sensor, it can be fixed at an angle of up to 30° to reduce the impact of air bubbles, sludge, etc. on the optical sensor window and to maintain a longer maintenance cycle.
- 3) The sensor is input installation, installation should be based on the specific circumstances of the site to use fixed clamps, clamps, conversion joints, etc. to fix the sensor, as far as possible to avoid the sensor bumping into hard objects, which will cause external damage, resulting in water leakage and other situations.
- 4) **Do not use the sensor's own cable to lift the sensor**, install cable protection devices to prevent the cable and the sensor between the taut tension is too large, resulting in cable breakage.
- 5) The depth of the water level at which the sensor is installed should not exceed 2 metres from the horizontal, and the depth of the water at the installation location should not be less than 30cm.
- 6) The sensor should not be installed with the measuring surface facing upwards, and should be installed vertically downwards or horizontally as far as possible.
- 7) If the sensor is installed in a circulation pool, drinking water tank or other environment, please contact the company's sales staff to customise the sensor according to the parameters of the sensor and to exclude interference with the sensor caused by external environment such as water flow and air bubbles.
- 8) The sensor contains sensitive optical components and electronic devices, the sensor should be installed to ensure that the sensor is not subject to violent mechanical impact.
- 9) The standard power supply for the sensor is 6-12V, if other voltage levels are required, communicate with the sales office in advance, do not use external power supplies other than 6-12V to connect the sensor, otherwise the internal chip of the sensor will be burnt.

VI. Maintenance Schedule and Method

Maintenance Schedule

Unlike the electrochemically-dissolved oxygen probe technology, the fluorine-dissolved oxygen probe does not consume oxygen and does not require frequent cleaning (except when applied to viscous liquids).

Maintenance tasks	Recommended maintenance frequency
Cleaning sensor	Clean every 30 days
Check sensor and fluorescent cap for damage	Check every 30 days
Replace the fluorescent cap	Change once a year
Calibrate the sensor (if required by the competent authority)	According to the maintenance schedule required by the competent authority

Note:

The maintenance frequency in the above table is only a recommendation. Please maintain the sensor according to the actual usage of the sensor. However, the replacement of the fluorescent cap is recommended to be replaced once a year.

Maintenance Methods

Sensor Maintenance

1) Sensor outer surface:

Clean the outer surface of the sensor with tap water. If there is still debris left, wipe it with a soft cloth. For some stubborn dirt, add some household washing liquid to the tap water to clean it.

2) The outer surface of the fluorescent cap:

Remove the protective cover at the front end of the sensor, rinse the dirt on the sensor window with clean water, and finally cover the cover; if you need to wipe, use a soft cloth and carefully wipe the force direction.

3) Fluor cap inner surface:

If water vapor or dust intrudes into the inside of the fluorescent cap, the cleaning steps are as follows:

- Remove the fluorescent cap
- Flush the inner surface of the fluorescent cap with tap water
- For fats containing fat and oil, wash with warm water with household washing solution
- Flush the inner surface of the fluorescent cap with deionized water
- Gently dry all surfaces with a clean lint-free cloth and place in a dry place to allow the water to completely evaporate.

4) Check the cable of the sensor:

The cable should not be tightened during normal operation, otherwise the

internal wire of the cable may be broken, causing the sensor to not work properly;

5) Check the sensor's outer casing for damage due to corrosion or other causes.

6) Fluorescent cap daily storage:

Put it in a protective cover with a moist sponge when not in use, so that the sensor stays wet for a long time. If the head of the sensor's fluorescent cap is dry for a long time, it will cause drift of the measurement result, so it needs to be immersed in water for 48 hours before continuing to work.

Notes:

- 1) Avoid exposing the inner surface of the fluorescent cap to sunlight.
- 2) Do not touch the fluorescent film with your hands.
- 3) Avoid any direct application of mechanical stress (pressure, scratches, etc.) to the fluorescent film during use.

Common Problems

Error	Possible causes	Solutions
The operation interface cannot be connected or do not display measurement results	Controller and cable connection error	Reconnect the controller and cables
	Cable failure	Please contact us
	The fluorescent cap is not tightened or damaged	Reinstall and tighten the fluorescent cap or replace the fluorescent cap
The measured value is too high, too low or the value is continuously unstable	The outer surface of the fluorescent cap is attached by foreign objects	Clean the outer surface of the fluorescent cap and agitate the probe during measurement
	Fluorescent cap is damaged	Replace the fluorescent cap
	The fluorescent cap has exceeded its service life	
Temperature measurement exceeds measurable range or garbled reading	Controller and cable connection error	Reconnect controller and temperature sensor cable
	Temperature sensor is attached to foreign objects	Gently brush the attachment with a soft brush

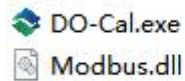
VII. Dissolved Oxygen Calibration

Automatic software calibration / manual calculation of calibration

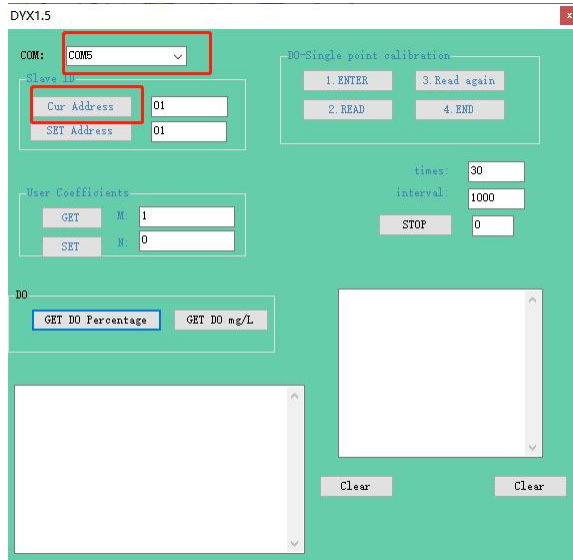
1. Automatic software calibration

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Before calibration, connect the sensor to the PC, remove the protective cap from the sensor and open the DO-Cal software.



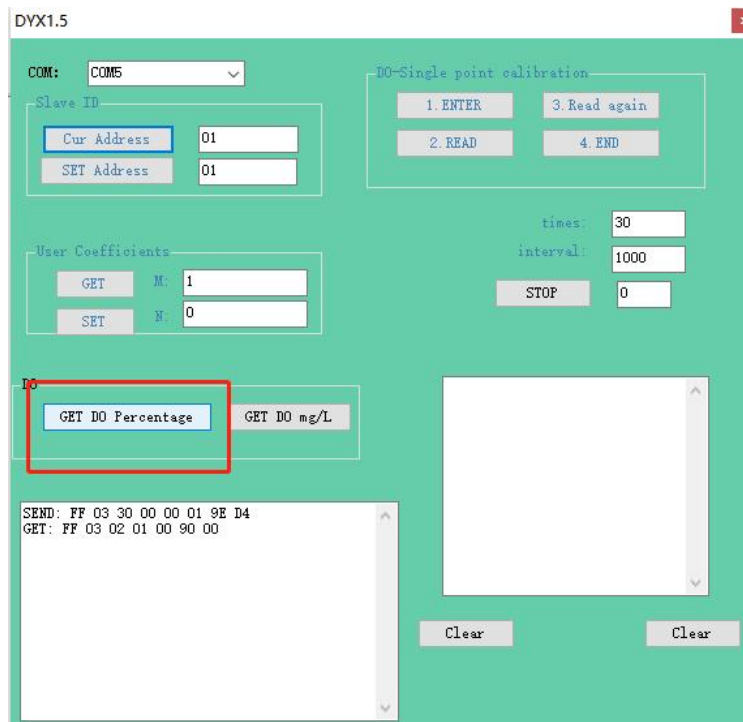
Step 1: Open the software and get the address

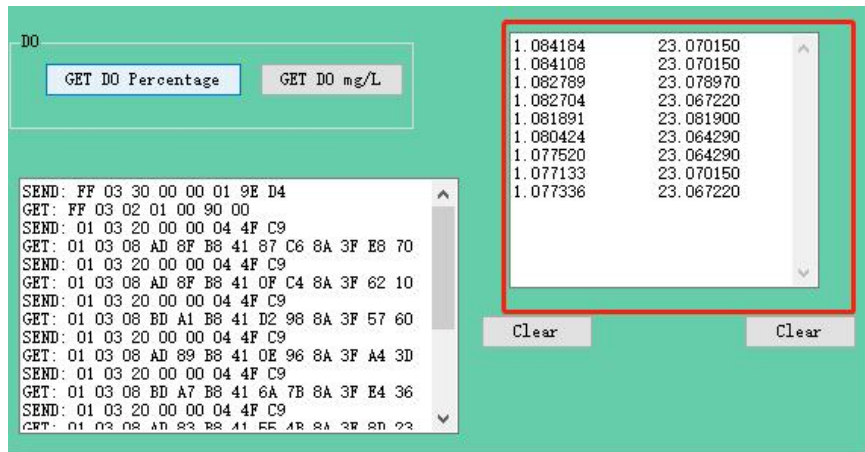


Step 2:

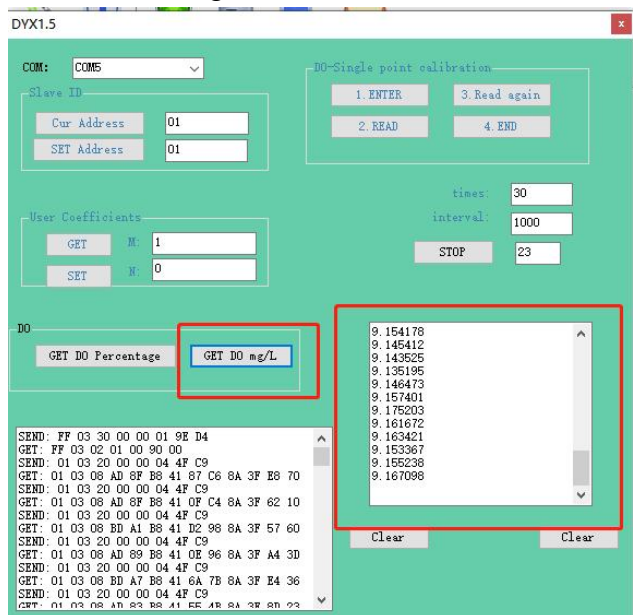
Before start calibration, let's test the sensor and see what data we get.

➤ Get DO Percentage

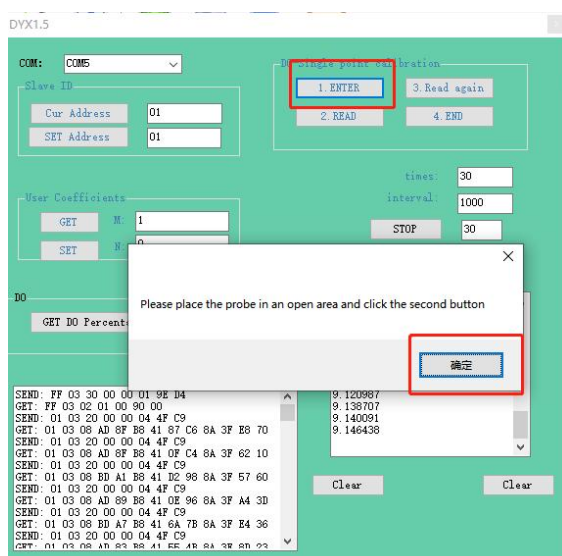




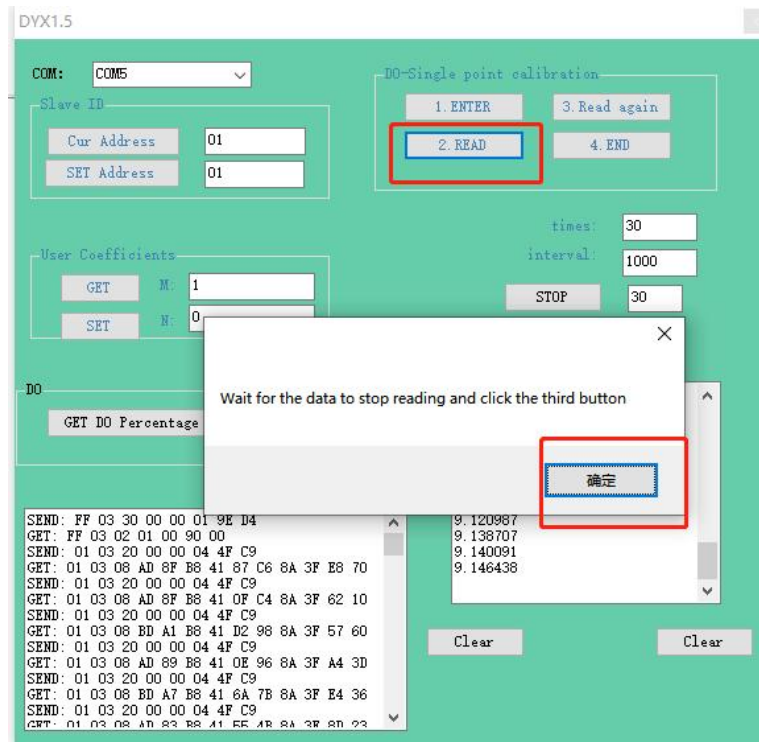
➤ Get DO mg/L



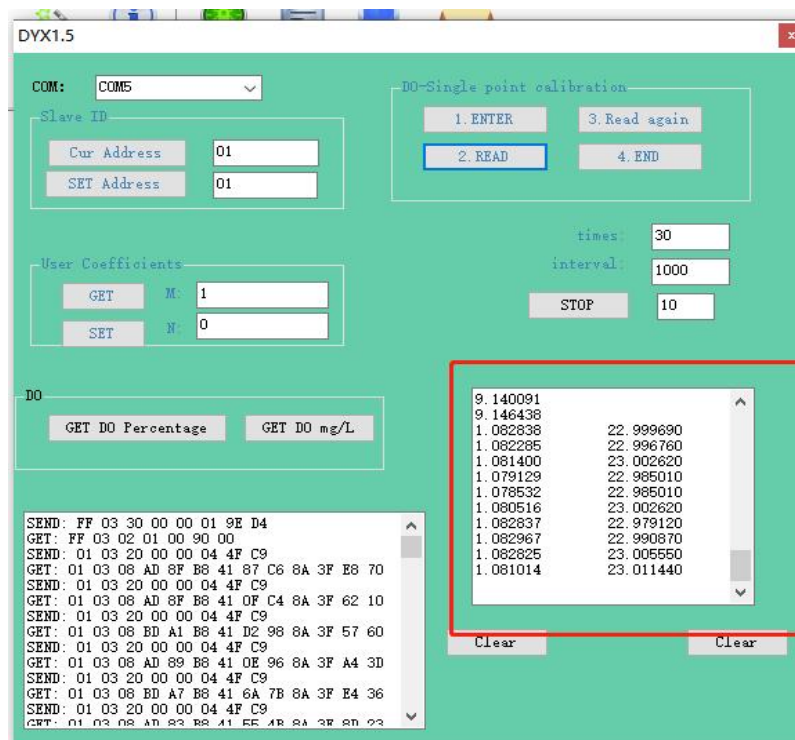
Step 3: Click the 1.ENTER



Step 4: Click the 1.READ

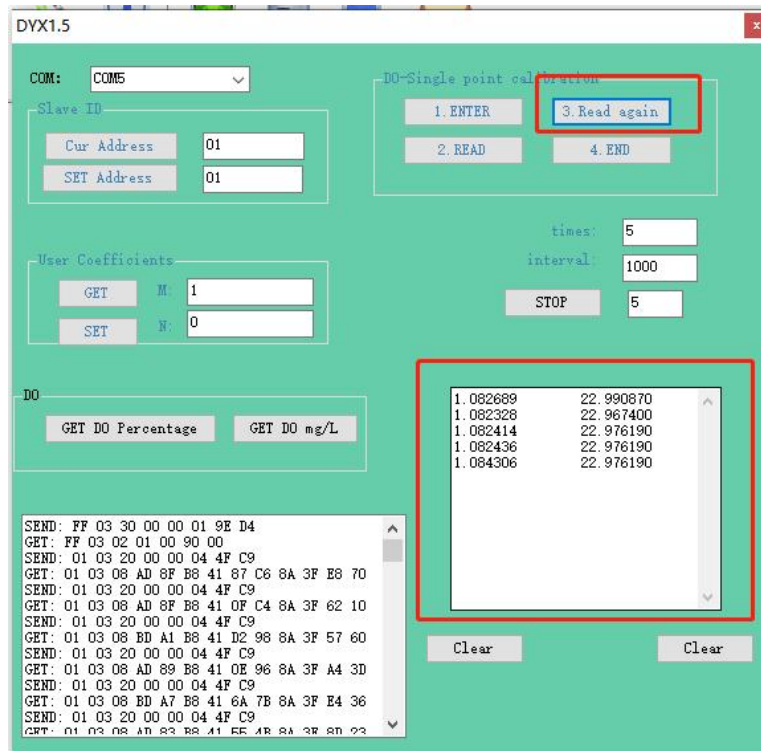


We will see the data here.

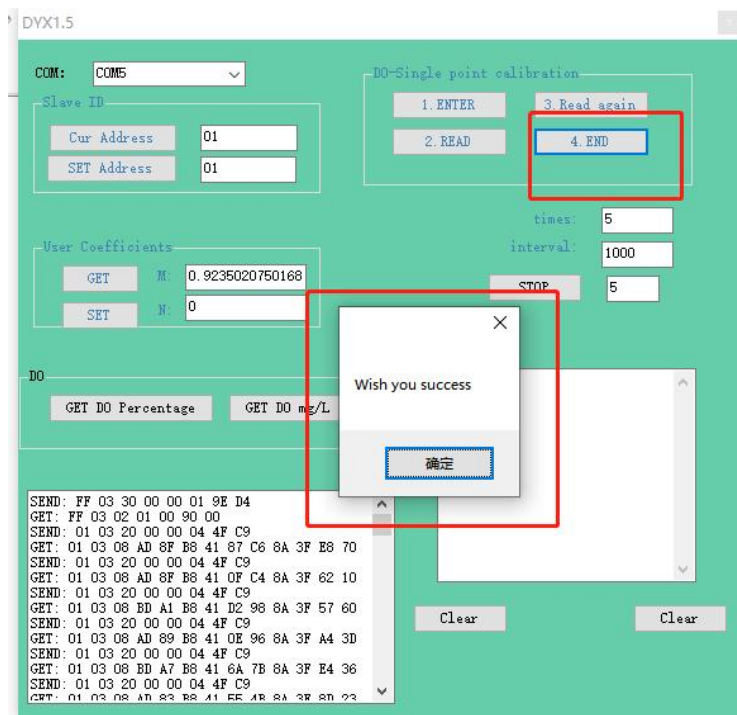


We will see this group of test data.

Step 5: Click the [3.READ again](#)



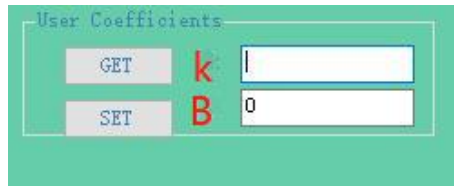
Step 6: Click the 4.END



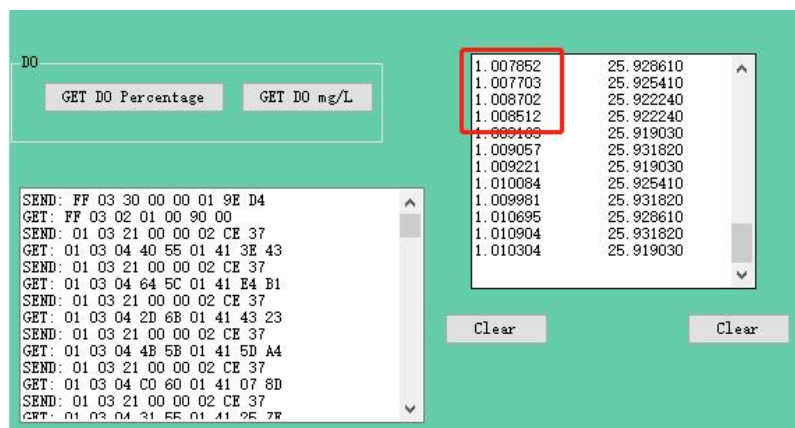
Calibration completed

2. Manual calculation of calibration

Change K&B



1. Normally we put the sensor in air and the percentage should be $1.00/(100\%)$
2. To calibrate, **only change k**
If the actual percentage we get is 1.12, then $1 \div 1.12 = 0.892857 \dots$, we get the value 0.892857
3. Our first step is to restore **k=1, B=0**
Then change the value of K to 0.89 and click **SET**
4. Calibration is completed.



Click **SET** for Save.

VIII. Quality Assurance

Dear Customer:

First of all, thank you for choosing to buy our products!

In order to guarantee the quality of our products, to clarify the responsibility of both the buyer and the seller for the quality of our products, to ensure the quality of our products and to guarantee the safety of our products, we would like to make the following guarantee.

The pH sensors provided by our company are guaranteed for one year from the date of delivery.

Xi'an Desun Uniwill Electronic Technology Co., Ltd.

Upon receipt of a product quality problem from a customer, we will confirm within 24 hours whether the product requires repair; do not send the product back without permission for repair.

In principle, the quality of the products provided by our company will be in accordance with the technical standards stipulated by the purchaser. If the purchaser does not specify the technical standards, we will follow the current national standards or implement the relevant factory standards, and ensure the stability and gradual improvement of product quality.

The supplier guarantees that the products supplied meet the statutory quality standards and is responsible for the quality of the products, and that the product packaging and registered trademarks comply with the relevant national regulations.

The purchaser is responsible for the quality of the products due to poor storage and maintenance of the products.

IX. Warranty limit

The warranty does not cover the following:

- Damage caused by natural disasters, riots, war and other force majeure factors.
- Damage caused by improper use, negligence, accidents or improper use by the user.
- Shipping costs incurred for the return of the product.
- Costs incurred for postage of product parts during the warranty period.
- Travel costs for out-of-warranty repairs.

The remedies such as repair, replacement, or refund of the payment as described above are special cases that do not violate this warranty. The remedies such as replacement or refund of the payment are for our products. Based on strict liability or other legal theories, our company is not responsible for any other damage caused by product defects or negligent operation, including the subsequent damage caused by causality with these situations.